

DiaSorin	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

SECTION 1 IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

1.1.1 Trade name/designation	1.1.2 Code
[CONJ] Conjugate	Kit components: LIAISON® MUREX HCV Ab [REF] 318240
[CONJ] Conjugate	Kit components: LIAISON® XL MUREX HIV Ab/Ag HT [REF] 318290

The information contained in this safety data sheet refers exclusively to the components listed in point 1.1.1.

1.2 Relevant identified uses of the substance or mixture, and uses advised against

Laboratory reagents for in vitro diagnostics. See package insert for instructions. Do not use products for purposes other than those designated. Improper use could result in uncalculated safety or exposure risks.

1.3 Details of the supplier of the safety data sheet

Manufacturer:	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR Tel: +44 (0) 844 225 8999
Responsible Person:	E-mail: MSDSHelp@Diasorin.co.uk
USA Distributor:	Diasorin Inc. 1951 Northwestern Avenue Stillwater, MN 55082 USA Tel: 651-439-9710

1.4 Emergency telephone number

Country USA National Poison Control Center	Telephone Number Telephone number: 800-222-1222
United Kingdom of Great Britain and Northern Ireland National Poisons Information Service (Birmingham Unit) City Hospital Dudley Rd Birmingham	Telephone number: +44 121 507 4123 Emergency telephone number: 844 892 0111 Fax: +44 121 507 55 88 E-mail: mail@npis.org
National Poisons Information Service Edinburgh Scottish Poisons Information Bureau Royal Infirmary 51 Little France Crescent Edinburgh	Telephone number: +44 131 242 1383 Emergency telephone number: 844 892 0111 Fax: +44 131 242 13 87 E-mail: spib@luht.scot.nhs.uk

SECTION 2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

REGULATION (EC) No 1272/2008	
STOT RE	H373
Skin sens. 1	H317

DiaSorin	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

2.2 Label elements

	REGULATION (EC) No 1272/2008
SIGNAL WORD:	Warning
SYMBOLS / PICTOGRAMS:	  GHS08 Health hazard GHS07 Exclamation mark
HAZARD STATEMENTS:	H373 May cause damage to organs through prolonged or repeated exposure. H317 May cause an allergic skin reaction.
PRECAUTIONARY STATEMENTS:	P261 Avoid breathing dust/fume/gas/mist/vaporous/spray. P280 Wear protective gloves/protective clothing/eye protection/face protection. P333+P313 If skin irritation or rash occurs: Get medical advice / attention. P362/P364 Take off contaminated clothing and wash it before reuse.

Contains: reaction mass of: 5-Chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H-isothiazol-3-one [EC no. 220-239-6] (3:1).

2.3 Other hazards

Potential biological hazard: contains biological materials of human/animal origin.

SECTION 3 COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substances

N.A

3.2 Mixtures

Mixture of: (chemical name of substances in mixture, (example: 5-chloro-2-methyl-2H-isothiazol-3-one [EC No. 247-500-7]; 2-methyl-2H-isothiazol-3-one [EC No. 220-239-6] (3:1)) .

CAS No.	CE No.	Index No.	Concentration	Classification Reg. 1272/2008 (pure substance)
55965-84-9	--	613-167-00-5	≤ 0.01 %	Acute Tox. 3 H331 Acute Tox. 3 H311 Acute Tox. 3 H301 Skin Corr. 1B H314 Skin Sens. 1 H317 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

Ethylene Glycol

CAS No.	CE No.	Index No.	Concentration	Classification Reg. 1272/2008
107-21-1	203-473-3	603-027-00-1	≤ 10 %	STOT RE 2 H373 Acut Tox. 4 H302

Refer to Section 16 for complete text of hazard indication and class.

SECTION 4 FIRST AID MEASURES

4.1 Description of first aid measures

	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

Seek medical advice for first aid; protect first aid providers; wear suitable hand and eye protection and, if necessary, breathing protection.

Inhalation: Take the person outdoors to fresh air and seek medical advice/attention immediately. If not breathing give artificial respiration. If breathing is difficult, give oxygen.

Skin contact: Remove contaminated clothing. Disinfect and wash skin with soap and water. Seek medical advice.

Eye contact: Rinse thoroughly with water or suitable eyewash. Seek medical advice.

Ingestion: Have the person drink as much water as possible. Do not induce vomiting. Seek medical advice.

4.2 Most important symptoms and effects, both acute and delayed

No information available.

4.3 Indication of any immediate medical attention and special treatment needed

Seek medical attention as soon as possible.

SECTION 5 FIREFIGHTING MEASURES

5.1 Extinguishing media

Use suitable extinguishing media, such as: carbon dioxide, foams, sprayed water, sand or aggregates.

5.2 Special hazards arising from the substance or mixture

Non-combustible preparation. Ambient fire may liberate hazardous vapours.

5.3 Advice for firefighters

Wear suitable protective clothing, including helmet and mask with self-contained breathing apparatus, CE certified according with Directive 89/686/EEC or NIOSH approved (USA).

Remove containers from the area of the fire if it is possible to do so without incurring risks.

Prevent fire-fighting water from entering surface water or groundwater. Collect in containers suitable for disposal.

SECTION 6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Free the contaminated area of non-emergency personnel.

Let fresh air enter contaminated area.

Avoid coming into contact with the substance and handling damaged containers without wearing suitable protective clothing and personal protection equipment for the face, hands and air-ways, CE certified according with Directive 89/686/EEC or NIOSH approved (USA).

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Prevent product from entering drains. Should not be released into the environment.

6.3 Methods and material for containment and cleaning up

Cover spillage with inert absorbent material.

Recover spilled material by mechanical means and collect in clean and suitably labelled containers.

Thoroughly wash and disinfect contaminated surfaces.

6.4 Reference to other sections

Dispose of recovered material as waste (see section 13); use personal protection equipment as indicated in section 8.

SECTION 7 HANDLING AND STORAGE

7.1 Precautions for safe handling

	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

Avoid contact with eyes, skin or clothing. Use in a suitably ventilated room. Do not eat, drink or smoke during use. Do not pipette by mouth. Wear lab coats, safety goggles and disposable gloves. Avoid splashing or forming an aerosol. Wash hands carefully on completion of operations.

7.2 Conditions for safe storage, including any incompatibilities

Store the preparation at 2-8°C hermetically sealed and away from the light. Keep away from foodstuffs, drinks and animal feed. Protect containers against damage.

7.3 Specific end uses

The mixtures are intended specifically for in vitro use, for the examination of samples of human origin. Handle in accordance with good laboratory practice, while also considering the risks deriving from the materials under analysis. Follow universal precaution guidelines for handling infectious agents and good laboratory practices such as those established by the US Centers for Disease Control and the National Institutes of Health, or the World Health Organization.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

No product information available;

Available information on hazardous components:

ETHYLENE GLYCOL (CAS No. 107-21-1)					
Country	Limit Value – 8 Hrs		Limit Value – Short Term		Legal Basis
	ppm	mg/m ³	Ppm	mg/m ³	
United States	-	-	-	-	OSHA – 29 CFR 1910 PEL
	-	-	50	125	ACGIH / NIOSH – Threshold Limit Values (TLVs) / Recommended Exposure Limits (RELs)

8.2 Exposure controls

Eyeface protection: wear safety goggles or visor; do not use contact lenses. Protect eyes with safety glasses or goggles based upon the hazards in the workplace and the potential exposure risks. Safety glasses and goggles should meet, at a minimum, the ANSI standard Z87.1 (US) / EN 166 (EU) or an equivalent level of protection.

Hand protection: wear safety goggles or visor; do not use contact lenses. Protect eyes with safety glasses or goggles based upon the hazards in the workplace and the potential exposure risks. Safety glasses and goggles should meet, at a minimum, the ANSI standard Z87.1 (US) / EN 166 (EU) or an equivalent level of protection.

Body protection: Prevent skin contact and protect clothing. Wear a lab coat to prevent skin contact.

Breathing protection: avoid inhaling aerosol. None required with normal use of products for diagnostics purposes and adequate ventilation conditions exist. The use of adequate ventilation must always take priority over PPE. If it is not possible to keep the concentration below the permitted workplace exposure thresholds, then respiratory protection must be used.

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

PROPERTY

APPEARANCE: LIQUID

ODOR: N.A

ODOR THRESHOLD: N.A

PH: 6 TO 8

MELTING POINT/FREEZING POINT: NO INFORMATION AVAILABLE

INITIAL BOILING POINT AND BOILING RANGE: NO INFORMATION AVAILABLE

FLASH POINT: > 60 °C

EVAPORATION RATE: NO INFORMATION AVAILABLE

	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

PROPERTY

FLAMMABILITY (SOLID, GAS): NO INFORMATION AVAILABLE

UPPER/LOWER FLAMMABILITY OR EXPLOSIVE LIMITS: NO INFORMATION AVAILABLE

VAPOR PRESSURE: NO INFORMATION AVAILABLE

VAPOR DENSITY: NO INFORMATION AVAILABLE

RELATIVE DENSITY: NO INFORMATION AVAILABLE

SOLUBILITY(IES): NO INFORMATION AVAILABLE

PARTITION COEFFICIENT (N-OCTANOL/WATER): NO INFORMATION AVAILABLE

AUTO-IGNITION TEMPERATURE: NO INFORMATION AVAILABLE

DECOMPOSITION TEMPERATURE: NO INFORMATION AVAILABLE

VISCOSITY: NO INFORMATION AVAILABLE

EXPLOSIVE PROPERTIES: NOT CLASSIFIED AS EXPLOSIVE

OXIDIZING PROPERTIES: NONE

9.2 Other information

No data available.

SECTION 10 STABILITY AND REACTIVITY

10.1 Reactivity

There are no particular risks of reaction with other substances in normal conditions of use.

10.2 Chemical stability

The product is chemically stable in normal conditions of use and storage.

10.3 Possibility of hazardous reactions

No hazardous reactions are foreseeable in normal conditions of use and storage.

10.4 Conditions to avoid

None in particular. However, the usual precautions used for chemical products should be respected.

10.5 Incompatible materials

Information not available.

10.6 Hazardous decomposition products

No information available

SECTION 11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

No product information available.

Available information on hazardous components:

Ethylene glycol

LD50 (oral), > 2000 mg/kg

reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H -isothiazol-3-one [EC no. 220-239-6] (3:1)

Acute toxicity by mouth, skin and inhalation

LD-50 (oral), rat: 60 mg/kg (active isothiazol)

LD-50 (dermal), rabbit: 80 mg/kg (active isothiazols)

LC-50 4-hour (inhalation), rat: 330 mg/m3 (active isothiazols)

	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

In humans, solutions containing more than 0.5% (5000 ppm) active isothiazols cause severe skin irritation and may be corrosive to the mucous membranes and cornea.

Corrosion, irritation and sensitization

The substance has manifested sensitizing effects.

In humans, solutions containing more than 0.5% (5000 ppm) active isothiazols cause severe skin irritation and may be corrosive to the mucous membranes and cornea.

The application of single doses of 0.5 ml of an aqueous solution (with active isothiazols at 560, 2800 and 5600 ppm) to the skin of rabbits, causes after 24 hours, severe skin damage at 5600 ppm and moderate damage at 2800 ppm; no effects are observed at 560 ppm.

Long-term toxicity (sub-acute, sub-chronic, chronic)

In a study conducted on rats, the inhalation of concentrations of active isothiazols of 0, 0.027, 0.23 and 0.89 mg/m³ in aerosol form, for 6 hours/day for 5 days/week for 13 weeks, caused, at the high doses, reduced body growth in both sexes, reduced blood proteins in females and reduced spleen weight in males. Histopathological examination revealed mild rhinitis at the dose of 0.23 mg/m³.

Carcinogenicity, Mutagenicity and Reproductive Toxicity (fertility and development)

In a study of carcinogenicity conducted on mice by means of cutaneous administration of substance, the substance did not manifest cutaneous and/or systemic carcinogenic potential.

Assays of mutagenicity showed the substance to be cytotoxic and directly mutagenic in vitro, in two species of bacteria and one mammalian cell line. In studies in vitro on mammalian cells it was not observed to cause damage to DNA, and in vivo, no formation of bonds with DNA was observed.

The substance manifested no reproductive toxicity.

SECTION 12 ECOLOGICAL INFORMATION

12.1 Toxicity

Mixture: no information available.

Available information on hazardous component:

Ethylene glycol

Parameter	Animal species	Time of exposure	Data
LC50	Oncorhynchus mykiss (rainbow trout)	96 hours	18,500 mg/l
EC50	Crustaceans	24 hours	74,000 mg/l

reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H -isothiazol-3-one [EC no. 220-239-6] (3:1); short-term effects:

Parameter	Animal species	Time of exposure	Data
LC50	Fish	96 hours	0.19 mg/l
EC50	Crustaceans	48 hours	0.16 mg/l

long-term effects: no information available.

12.2 Persistence and degradability

No information available.

12.3 Bioaccumulative potential

No information available

12.4 Mobility in soil

If spilled, may enter ground.

12.5 Results of PBT and vPvB assessment

No information available.

12.6 Other adverse effects

No information available.

DiaSorin	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

SECTION 13 DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Dispose of in accordance with all federal, state and local environmental regulations.

SECTION 14 TRANSPORT INFORMATION

The mixture is not to be considered hazardous within the meaning of the provisions in force governing the transport of hazardous materials by road (ADR), rail (RID), sea (IMDG code) and air (ICAO, IATA); transport at 2-8°C.

14.1 UN number: N.A.

14.2 UN proper shipping name: N.A.

14.3 Transport hazard class(es): N.A.

14.4 Packing group: N.A.

14.5 Environmental hazards: N.A.

14.6 Special precautions for user: N.A.

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code: N.A.

SECTION 15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Canada	This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR.	
	DSL:	Y
	NDSL:	N
EU	Classification and procedures used to derive the classification for products listed in Section 1.1 according to European Union regulations and directives: 67/548/EEC, 1999/45/EC, 2001/60/EC and Regulation (EC) 1272/2008. This MSDS complies with regulations on Safety Data Sheet preparation; Directive 2001/58/CE and REACH Regulation 1907/2006/CE, article 31.	
	Restriction pursuant to Annex XVII to EC Regulation 1907/2006	Point 3
	Seveso category	none
	Candidate List (art. 59 Regulation 1907/2006)	NA
	authorization (Annex XIV REACH)	NA
	Water Hazard Class (Germany):	WGK 1
	Rotterdam Convention - annex III	NA
United States	Product components listed in Section 1.1 are classified in accordance with OSHA Hazard Communication Standard, 29 CFR 1910.1200, and individual US state Right-to-Know laws.	
	CAA:	N – Does not contain any hazardous air pollutants (HAPs) or Class 1 or 2 Ozone depleters.
	CWA:	N
	CERCLA:	RQ – 1,000 lbs.
	SARA 313:	Y – Releases > 1.0 lbs into the air, water or land must be reported.
	TSCA:	Y
	CA Prop 65:	N

15.2 Chemical safety assessment

No chemical safety assessment has been produced for the mixture and the substances contained therein.

SECTION 16 OTHER INFORMATION

Text of hazard (H) indications mentioned in section 2-3 of the sheet:

Hazard statements and class

	DiaSorin Italia S.p.A. UK Branch Central Road Dartford, DA1 5LR www.diasorin.co.uk	SAFETY DATA SHEET MSDS_en_K_KMSDS52
Language	en	
Revision No.:	03	
Date:	2024-01-16	

Acute Tox.: Acute toxicity
Skin Corr.: Skin corrosion
Skin Sens.: Skin sensitization
Aquatic Acute: Hazardous to the aquatic environment, acute toxicity
Aquatic Chronic: Hazardous to the aquatic environment, chronic toxicity
H373 Specific target organ toxicity – repeated exposure, Category 2
H331 Toxic if inhaled
H311 Toxic in contact with skin
H302 Acute toxicity
H301 Toxic if swallowed
H314 Causes severe skin burns and eye damage
H317 May cause an allergic skin reaction
H400 Very toxic to aquatic life
H410 Very toxic to aquatic life with long-lasting effects

Text of precautionary (P) statements mentioned in section 2-3 of the sheet:

P261 Avoid breathing dust/fume/gas/mist/vaporous/spray.

P280 Wear protective gloves / protective clothing / eye protection / face protection

P333+P313 If skin irritation or rash occurs: Get medical advice/attention.

P362/P364 Take off contaminated clothing and wash it before reuse.

GENERAL BIBLIOGRAPHY

1. Regulation (EC) 1907/2006 (REACH) of the European Parliament
2. Regulation (EC) 1272/2008 (CLP) of the European Parliament
3. The Merck Index. - 10th Edition
4. Handling Chemical Safety
5. CDC - NIOSH website – Guide to Chemical Hazards
6. INRS - Fiche Toxicologique (toxicological sheet)
7. Patty - Industrial Hygiene and Toxicology
8. N.I. Sax - Dangerous properties of Industrial Materials-7, 1989 Edition
9. ECHA website
10. US OSHA 29 CFR 1910.1200
11. eChemPortal website
12. IARC website – Evaluation of carcinogenic risks to humans
13. Health Canada website
14. Canada Labour Health and Safety website
15. CCOHS website
16. World Health Organization - Laboratory biosafety manual , 3rd ed.

All information is correct, to the best of our knowledge, on the date of issue of the data sheet. It is provided for information purposes only, however, and does not constitute a guarantee.

Procedures for use: see instructions in the package. Do not use the product for any purpose other than that for which it is intended.

Judgments as to the suitability of information herein for the purchaser's purposes are necessarily the purchaser's responsibility.

Although reasonable care has been taken in the preparation of such information, the company extends no warranties, makes no representations and assumes no responsibility as to accuracy or suitability of such information for application to purchaser's intended purposes and accepts no responsibility for any injury, loss or damage deriving from improper use of the product.

The product is employed under its users' control and it is their responsibility to comply with the correct operating procedures indicated, as well as to observe proper laboratory hygiene.